

# Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

## References

- [jqassistant](#)
- [Neo4j Python Driver](#)

## Node Labels

### Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

|    | nodeLabels                                        | nodesWithThatLabels | nodesWithThatLabelsPercent |
|----|---------------------------------------------------|---------------------|----------------------------|
| 0  | [Git, Update, Change]                             | 139893              | 46.017434                  |
| 1  | [Git, Change, Create]                             | 45414               | 14.938816                  |
| 2  | [Git, Change, Delete]                             | 22797               | 7.499013                   |
| 3  | [Java, ByteCode, Member, Method]                  | 13810               | 4.542763                   |
| 4  | [Java, ByteCode, Parameter]                       | 13459               | 4.427303                   |
| 5  | [Git, Commit]                                     | 11609               | 3.818750                   |
| 6  | [File, Git]                                       | 8332                | 2.740789                   |
| 7  | [Java, ByteCode, Bound, ParameterizedType]        | 7323                | 2.408882                   |
| 8  | [Java, ByteCode, Bound]                           | 7320                | 2.407895                   |
| 9  | [Git, Change, Rename]                             | 7018                | 2.308553                   |
| 10 | [Java, ByteCode, Member, Field]                   | 3724                | 1.225000                   |
| 11 | [Java, Value, ByteCode, Annotation]               | 2958                | 0.973026                   |
| 12 | [Java, ByteCode, Bound, WildcardType]             | 2950                | 0.970395                   |
| 13 | [Java, ByteCode, Member, Method, Constructor]     | 2177                | 0.716118                   |
| 14 | [Xml, Element]                                    | 2162                | 0.711184                   |
| 15 | [Xml, Text]                                       | 1450                | 0.476974                   |
| 16 | [Java, ByteCode, TypeVariable, Bound]             | 1112                | 0.365789                   |
| 17 | [Java, ByteCode, Member, Method, Lambda]          | 986                 | 0.324342                   |
| 18 | [Type, File, Java, ByteCode, ResolvedDuplicate... | 903                 | 0.297039                   |
| 19 | [Type, File, Java, Class, ByteCode]               | 869                 | 0.285855                   |
| 20 | [Json, Key]                                       | 750                 | 0.246711                   |
| 21 | [Value, Json, Scalar]                             | 734                 | 0.241447                   |
| 22 | [Java, Value, ByteCode, Primitive]                | 686                 | 0.225658                   |
| 23 | [Type, File, Java, ByteCode, JavaType]            | 659                 | 0.216776                   |
| 24 | [Java, GenericDeclaration, ByteCode, Member, M... | 579                 | 0.190461                   |
| 25 | [Type, File, Java, ByteCode, ExternalType]        | 419                 | 0.137829                   |
| 26 | [Git, Change, Copy]                               | 298                 | 0.098026                   |
| 27 | [Author, Git, Person]                             | 297                 | 0.097697                   |
| 28 | [Value, Array]                                    | 294                 | 0.096711                   |
| 29 | [Committer, Git, Person]                          | 243                 | 0.079934                   |

## Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

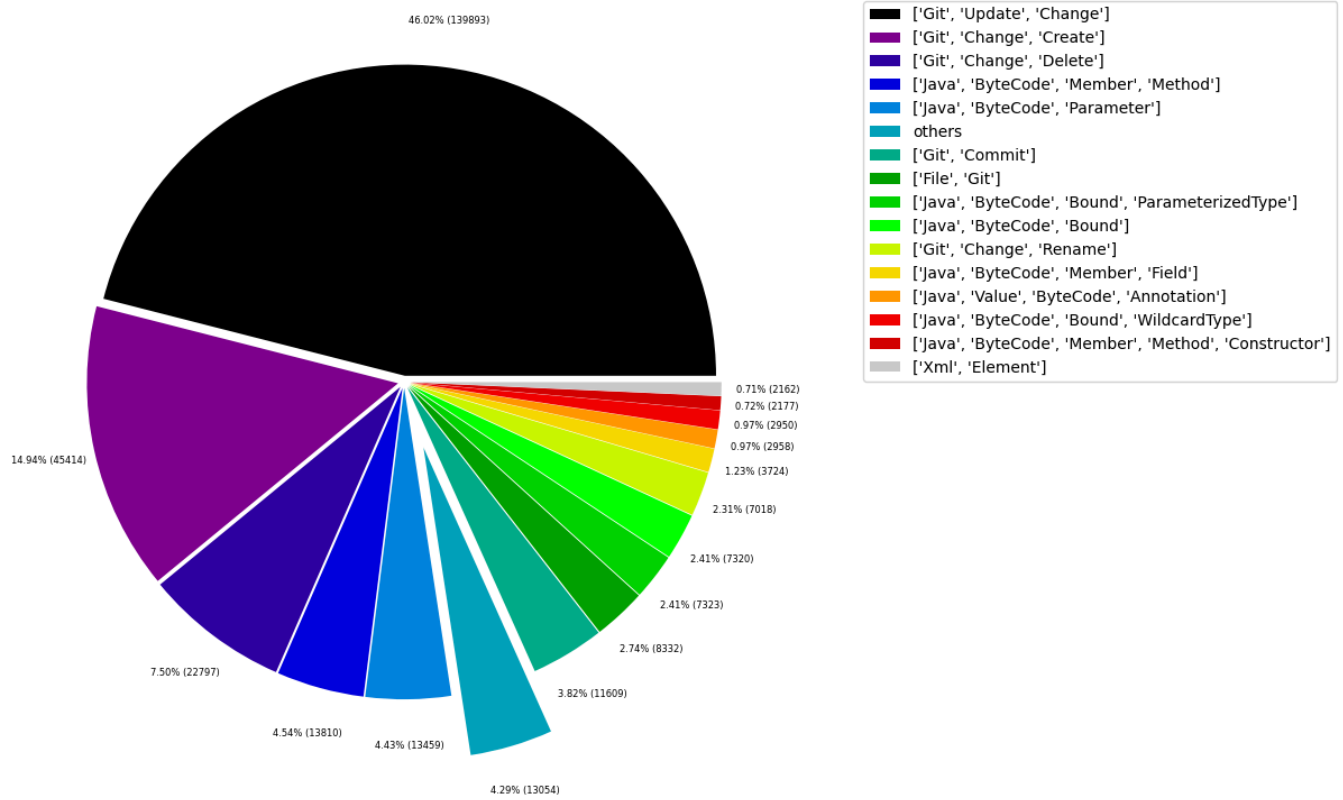


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

|    | nodeLabels                                        | nodesWithThatLabels | nodesWithThatLabelsPercent |
|----|---------------------------------------------------|---------------------|----------------------------|
| 0  | [Analyze, Task, jQAssistant]                      | 1                   | 0.000329                   |
| 1  | [Git, Branch]                                     | 1                   | 0.000329                   |
| 2  | [Repository, File, Git]                           | 1                   | 0.000329                   |
| 3  | [File, Json]                                      | 2                   | 0.000658                   |
| 4  | [File]                                            | 3                   | 0.000987                   |
| 5  | [Java, GenericDeclaration, ByteCode, Member, M... | 4                   | 0.001316                   |
| 6  | [Maven, Exclusion]                                | 5                   | 0.001645                   |
| 7  | [Value, Array, Json]                              | 6                   | 0.001974                   |
| 8  | [File, Maven, Xml, Pom, Document]                 | 9                   | 0.002961                   |
| 9  | [Type, File, Java, ByteCode, Void]                | 9                   | 0.002961                   |
| 10 | [Java, ManifestSection]                           | 9                   | 0.002961                   |
| 11 | [File, Artifact, Jar, Archive, Zip, Java]         | 9                   | 0.002961                   |
| 12 | [File, Java, Manifest]                            | 9                   | 0.002961                   |
| 13 | [File, Java, ServiceLoader]                       | 11                  | 0.003618                   |
| 14 | [File, Java, Properties]                          | 12                  | 0.003947                   |
| 15 | [Maven, ExecutionGoal]                            | 16                  | 0.005263                   |
| 16 | [Maven, PluginExecution]                          | 16                  | 0.005263                   |
| 17 | [Xml, Attribute]                                  | 18                  | 0.005921                   |
| 18 | [jQAssistant, Rule, Concept]                      | 19                  | 0.006250                   |
| 19 | [Type, File, Java, ByteCode, Throwable, Extern... | 21                  | 0.006908                   |
| 20 | [Maven, Plugin]                                   | 21                  | 0.006908                   |
| 21 | [Maven, Configuration]                            | 21                  | 0.006908                   |
| 22 | [Type, File, Java, ByteCode, Throwable, Resolv... | 22                  | 0.007237                   |
| 23 | [Type, File, Java, ByteCode, Enum]                | 30                  | 0.009868                   |
| 24 | [Type, File, Java, ByteCode, PrimitiveType]       | 31                  | 0.010197                   |
| 25 | [Xml, Namespace]                                  | 36                  | 0.011842                   |
| 26 | [Type, File, Java, ByteCode, Annotation]          | 44                  | 0.014474                   |
| 27 | [File, Directory]                                 | 45                  | 0.014803                   |
| 28 | [Type, File, Java, Class, ByteCode, Throwable]    | 56                  | 0.018421                   |
| 29 | [Type, File, Java, GenericDeclaration, ByteCod... | 85                  | 0.027961                   |

## Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (less than 0.5% overall)

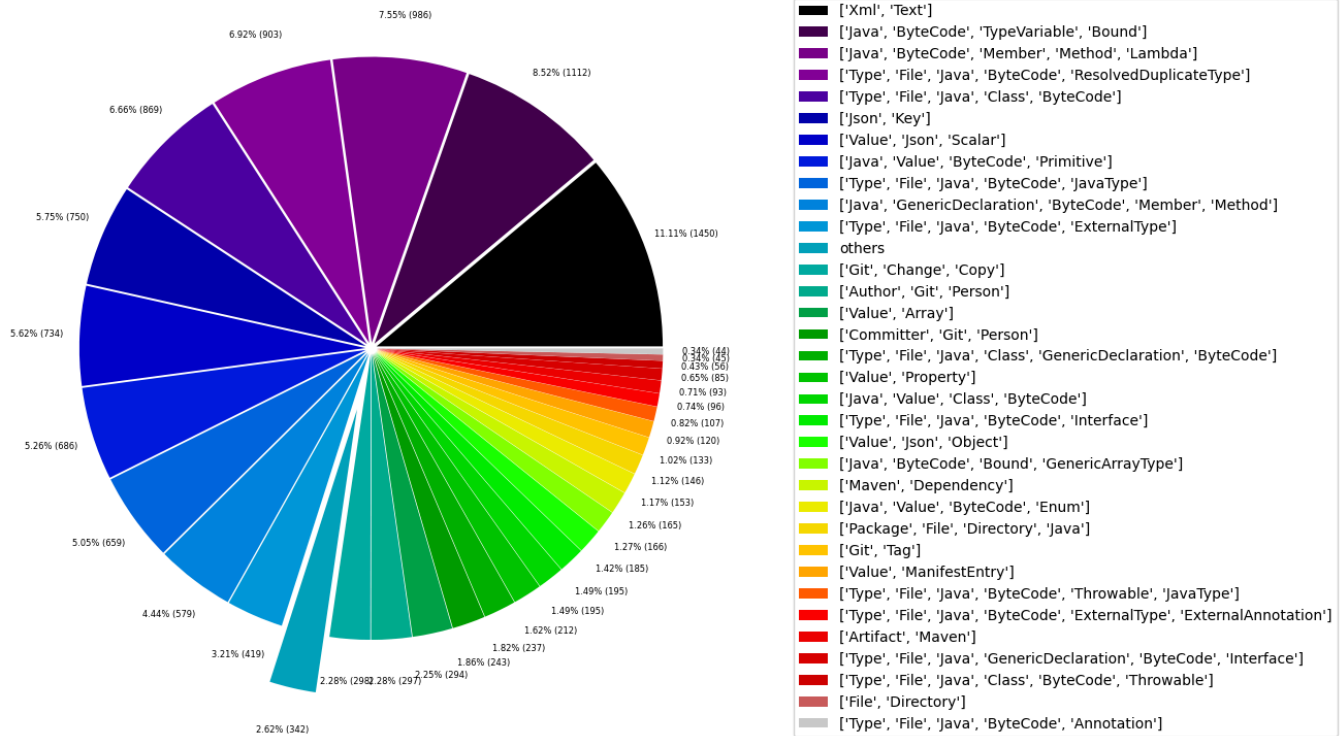


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

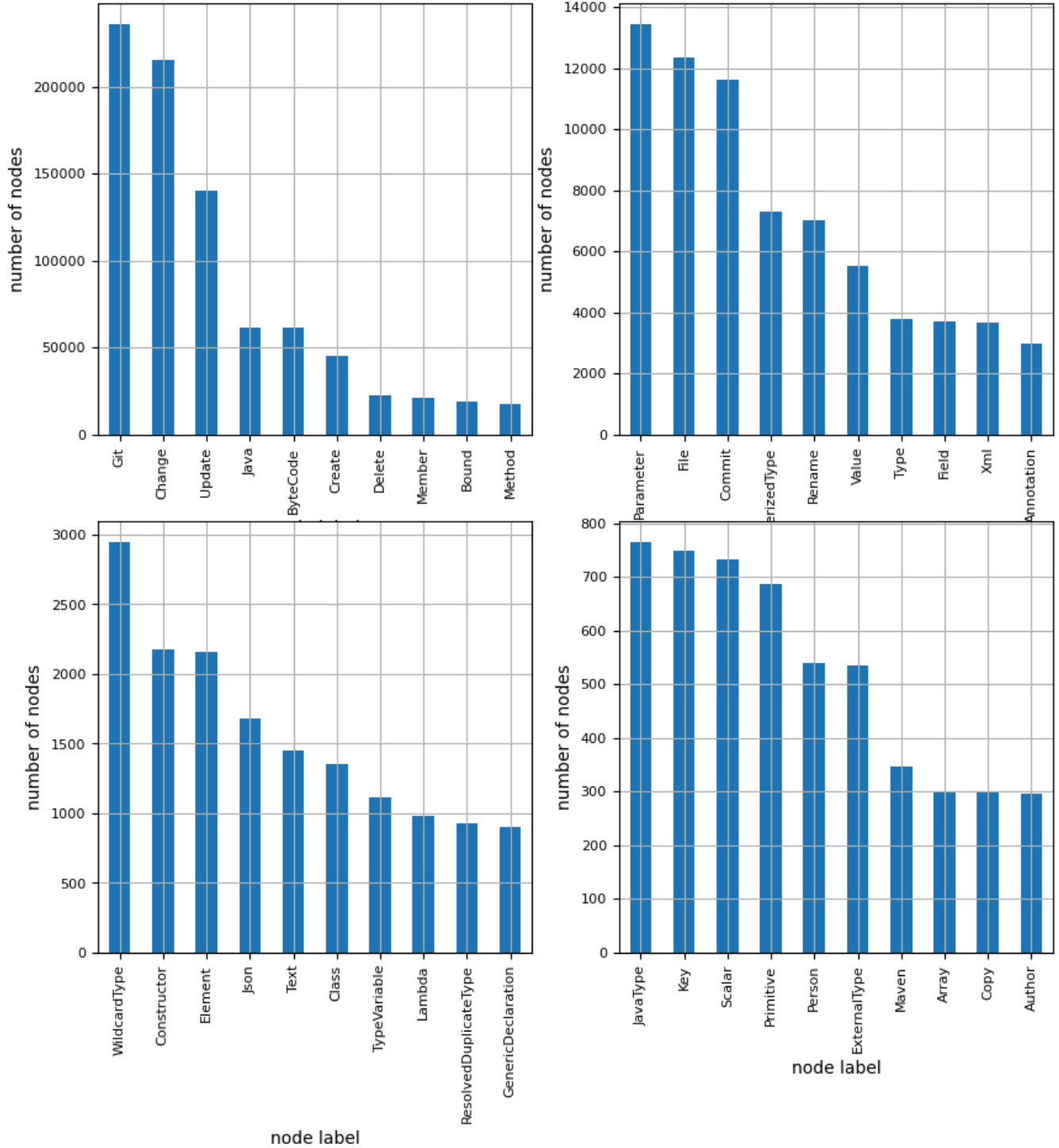
|    | nodeLabel             | nodesWithThatLabel | nodesWithThatLabelPercent |
|----|-----------------------|--------------------|---------------------------|
| 0  | Git                   | 236036             | 77.643421                 |
| 1  | Change                | 215420             | 70.861842                 |
| 2  | Update                | 139893             | 46.017434                 |
| 3  | Java                  | 61581              | 20.256908                 |
| 4  | ByteCode              | 61385              | 20.192434                 |
| 5  | Create                | 45414              | 14.938816                 |
| 6  | Delete                | 22797              | 7.499013                  |
| 7  | Member                | 21280              | 7.000000                  |
| 8  | Bound                 | 18871              | 6.207566                  |
| 9  | Method                | 17556              | 5.775000                  |
| 10 | Parameter             | 13459              | 4.427303                  |
| 11 | File                  | 12362              | 4.066447                  |
| 12 | Commit                | 11609              | 3.818750                  |
| 13 | ParameterizedType     | 7323               | 2.408882                  |
| 14 | Rename                | 7018               | 2.308553                  |
| 15 | Value                 | 5543               | 1.823355                  |
| 16 | Type                  | 3783               | 1.244408                  |
| 17 | Field                 | 3724               | 1.225000                  |
| 18 | Xml                   | 3675               | 1.208882                  |
| 19 | Annotation            | 3002               | 0.987500                  |
| 20 | WildcardType          | 2950               | 0.970395                  |
| 21 | Constructor           | 2181               | 0.717434                  |
| 22 | Element               | 2162               | 0.711184                  |
| 23 | Json                  | 1677               | 0.551645                  |
| 24 | Text                  | 1450               | 0.476974                  |
| 25 | Class                 | 1357               | 0.446382                  |
| 26 | TypeVariable          | 1112               | 0.365789                  |
| 27 | Lambda                | 986                | 0.324342                  |
| 28 | ResolvedDuplicateType | 925                | 0.304276                  |
| 29 | GenericDeclaration    | 905                | 0.297697                  |
| 30 | JavaType              | 766                | 0.251974                  |
| 31 | Key                   | 750                | 0.246711                  |
| 32 | Scalar                | 734                | 0.241447                  |
| 33 | Primitive             | 686                | 0.225658                  |
| 34 | Person                | 540                | 0.177632                  |
| 35 | ExternalType          | 536                | 0.176316                  |
| 36 | Maven                 | 346                | 0.113816                  |
| 37 | Array                 | 300                | 0.098684                  |
| 38 | Copy                  | 298                | 0.098026                  |
| 39 | Author                | 297                | 0.097697                  |

## Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

<Figure size 640x480 with 0 Axes>

## Node count by label



## Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 968646

|    | relationshipType         | nodesWithThatRelationshipType | nodesWithThatRelationshipTypePercent |
|----|--------------------------|-------------------------------|--------------------------------------|
| 0  | CONTAINS_CHANGE          | 215420                        | 22.239291                            |
| 1  | MODIFIES                 | 215420                        | 22.239291                            |
| 2  | UPDATES                  | 139893                        | 14.442118                            |
| 3  | CREATES                  | 52730                         | 5.443681                             |
| 4  | INVOKES                  | 37640                         | 3.885837                             |
| 5  | DELETES                  | 29815                         | 3.078008                             |
| 6  | CHANGED_TOGETHER_WITH    | 26210                         | 2.705839                             |
| 7  | COMMITTED                | 23218                         | 2.396954                             |
| 8  | DEPENDS_ON               | 22918                         | 2.365983                             |
| 9  | OF_TYPE                  | 22230                         | 2.294956                             |
| 10 | DECLARES                 | 21745                         | 2.244886                             |
| 11 | OF_RAW_TYPE              | 17483                         | 1.804891                             |
| 12 | HAS                      | 14616                         | 1.508910                             |
| 13 | HAS_PARENT               | 14206                         | 1.466583                             |
| 14 | RETURNS                  | 13084                         | 1.350751                             |
| 15 | HAS_COMMIT               | 11609                         | 1.198477                             |
| 16 | READS                    | 9705                          | 1.001914                             |
| 17 | HAS_ACTUAL_TYPE_ARGUMENT | 8459                          | 0.873281                             |
| 18 | HAS_FILE                 | 8332                          | 0.860170                             |
| 19 | RENAMES                  | 7018                          | 0.724516                             |
| 20 | OF_GENERIC_TYPE          | 6045                          | 0.624067                             |
| 21 | RESOLVES_TO              | 4308                          | 0.444745                             |
| 22 | SIMILAR                  | 4133                          | 0.426678                             |
| 23 | WRITES                   | 4053                          | 0.418419                             |
| 24 | CONTAINS                 | 4010                          | 0.413980                             |
| 25 | HAS_NEW_NAME             | 3662                          | 0.378053                             |
| 26 | RETURNS_GENERIC          | 3616                          | 0.373305                             |
| 27 | ANNOTATED_BY             | 2946                          | 0.304136                             |
| 28 | REQUIRES                 | 2267                          | 0.234038                             |
| 29 | HAS_FIRST_CHILD          | 2162                          | 0.223198                             |

## Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

<Figure size 640x480 with 0 Axes>



Relationship types (more than 0.5% overall)

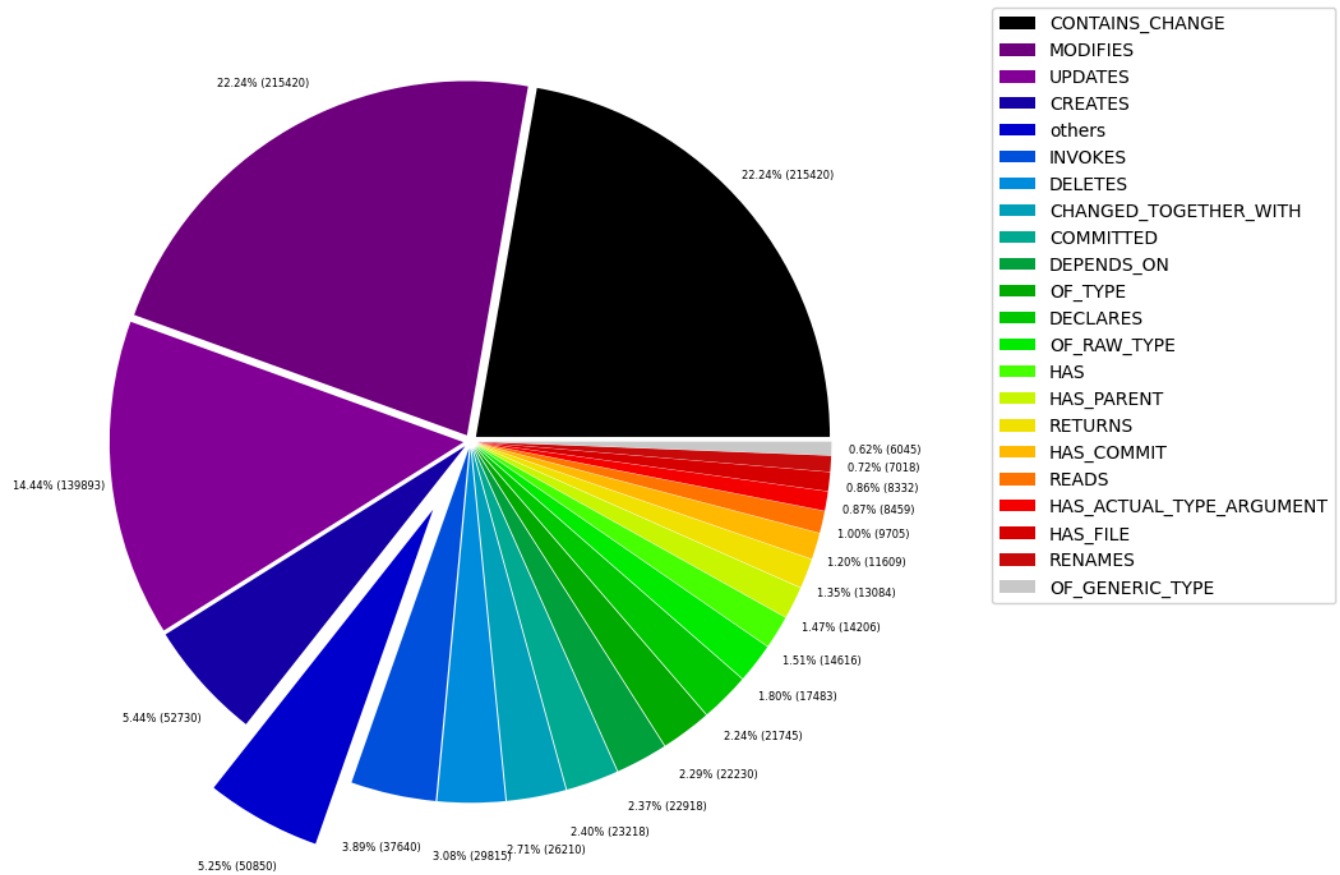


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

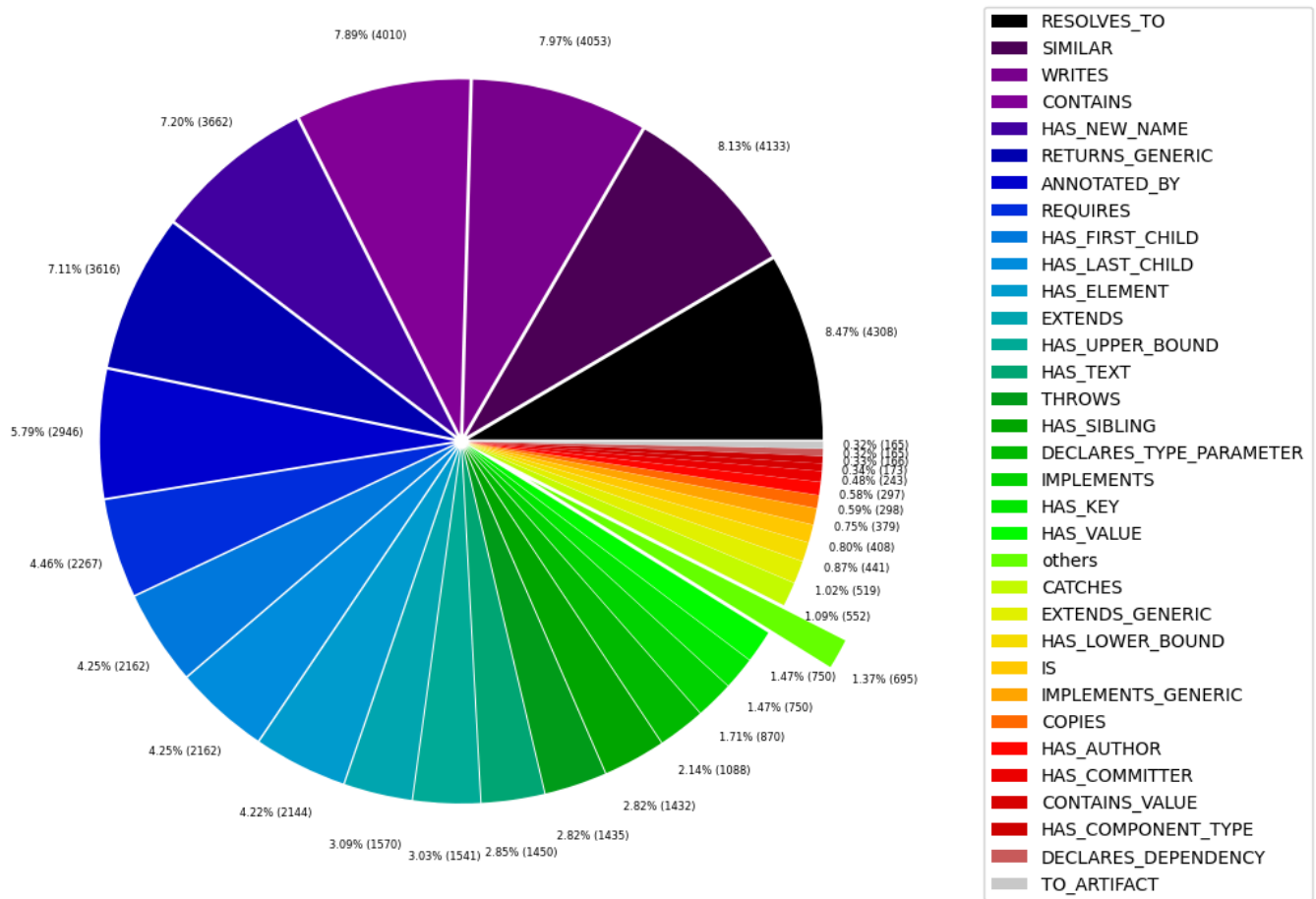
|    | relationshipType        | nodesWithThatRelationshipType | nodesWithThatRelationshipTypePercent |
|----|-------------------------|-------------------------------|--------------------------------------|
| 0  | HAS_PROPERTY            | 1                             | 0.000103                             |
| 1  | HAS_BRANCH              | 1                             | 0.000103                             |
| 2  | HAS_HEAD                | 2                             | 0.000206                             |
| 3  | THROWS_GENERIC          | 5                             | 0.000516                             |
| 4  | EXCLUDES                | 5                             | 0.000516                             |
| 5  | DESCRIBES               | 9                             | 0.000929                             |
| 6  | HAS_ROOT_ELEMENT        | 10                            | 0.001032                             |
| 7  | HAS_GOAL                | 16                            | 0.001652                             |
| 8  | HAS_EXECUTION           | 16                            | 0.001652                             |
| 9  | OF_NAMESPACE            | 18                            | 0.001858                             |
| 10 | HAS_ATTRIBUTE           | 18                            | 0.001858                             |
| 11 | INCLUDES_CONCEPT        | 19                            | 0.001962                             |
| 12 | USES_PLUGIN             | 21                            | 0.002168                             |
| 13 | IS_ARTIFACT             | 21                            | 0.002168                             |
| 14 | HAS_CONFIGURATION       | 21                            | 0.002168                             |
| 15 | REQUIRES_TYPE_PARAMETER | 24                            | 0.002478                             |
| 16 | REQUIRES_CONCEPT        | 28                            | 0.002891                             |
| 17 | DECLARES_NAMESPACE      | 36                            | 0.003717                             |
| 18 | HAS_DEFAULT             | 39                            | 0.004026                             |
| 19 | COPY_OF                 | 119                           | 0.012285                             |
| 20 | HAS_TAG                 | 133                           | 0.013731                             |
| 21 | ON_COMMIT               | 133                           | 0.013731                             |
| 22 | DECLARES_DEPENDENCY     | 165                           | 0.017034                             |
| 23 | TO_ARTIFACT             | 165                           | 0.017034                             |
| 24 | HAS_COMPONENT_TYPE      | 166                           | 0.017137                             |
| 25 | CONTAINS_VALUE          | 173                           | 0.017860                             |
| 26 | HAS_COMMITTER           | 243                           | 0.025087                             |
| 27 | HAS_AUTHOR              | 297                           | 0.030661                             |
| 28 | COPIES                  | 298                           | 0.030765                             |
| 29 | IMPLEMENTS_GENERIC      | 379                           | 0.039127                             |

## Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



## Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

|    | sourceLabels                                  | relationType             | targetLabels                               | numberOfRelationships | numberOfNodesWithSameLabelsAsSource | numbe |
|----|-----------------------------------------------|--------------------------|--------------------------------------------|-----------------------|-------------------------------------|-------|
| 0  | [Git, Commit]                                 | CONTAINS_CHANGE          | [Git, Update, Change]                      | 139893                | 11609                               |       |
| 1  | [Git, Update, Change]                         | MODIFIES                 | [File, Git]                                | 139893                | 139893                              |       |
| 2  | [Git, Update, Change]                         | UPDATES                  | [File, Git]                                | 139893                | 139893                              |       |
| 3  | [Git, Commit]                                 | CONTAINS_CHANGE          | [Git, Change, Create]                      | 45414                 | 11609                               |       |
| 4  | [Git, Change, Create]                         | CREATES                  | [File, Git]                                | 45414                 | 45414                               |       |
| 5  | [Git, Change, Create]                         | MODIFIES                 | [File, Git]                                | 45414                 | 45414                               |       |
| 6  | [Java, ByteCode, Member, Method]              | INVOKES                  | [Java, ByteCode, Member, Method]           | 22961                 | 13709                               |       |
| 7  | [Git, Commit]                                 | CONTAINS_CHANGE          | [Git, Change, Delete]                      | 22797                 | 11609                               |       |
| 8  | [Git, Change, Delete]                         | DELETES                  | [File, Git]                                | 22797                 | 22797                               |       |
| 9  | [Git, Change, Delete]                         | MODIFIES                 | [File, Git]                                | 22797                 | 22797                               |       |
| 10 | [File, Git]                                   | CHANGED_TOGETHER_WITH    | [File, Git]                                | 20075                 | 8332                                |       |
| 11 | [Git, Commit]                                 | HAS_PARENT               | [Git, Commit]                              | 14197                 | 11609                               |       |
| 12 | [Repository, File, Git]                       | HAS_COMMIT               | [Git, Commit]                              | 11609                 | 1                                   |       |
| 13 | [Committer, Git, Person]                      | COMMITTED                | [Git, Commit]                              | 11609                 | 243                                 |       |
| 14 | [Author, Git, Person]                         | COMMITTED                | [Git, Commit]                              | 11609                 | 297                                 |       |
| 15 | [Java, ByteCode, Member, Method]              | READS                    | [Java, ByteCode, Member, Field]            | 8729                  | 13709                               |       |
| 16 | [Java, ByteCode, Member, Method]              | HAS                      | [Java, ByteCode, Parameter]                | 8601                  | 13709                               |       |
| 17 | [Repository, File, Git]                       | HAS_FILE                 | [File, Git]                                | 8332                  | 1                                   |       |
| 18 | [Git, Commit]                                 | CONTAINS_CHANGE          | [Git, Change, Rename]                      | 7018                  | 11609                               |       |
| 19 | [Git, Change, Rename]                         | DELETES                  | [File, Git]                                | 7018                  | 7018                                |       |
| 20 | [Git, Change, Rename]                         | MODIFIES                 | [File, Git]                                | 7018                  | 7018                                |       |
| 21 | [Git, Change, Rename]                         | RENAMES                  | [File, Git]                                | 7018                  | 7018                                |       |
| 22 | [Git, Change, Rename]                         | CREATES                  | [File, Git]                                | 7018                  | 7018                                |       |
| 23 | [Java, ByteCode, Parameter]                   | OF_TYPE                  | [Type, File, Java, ByteCode, javaType]     | 6243                  | 13459                               |       |
| 24 | [File, Git]                                   | HAS_NEW_NAME             | [File, Git]                                | 3662                  | 8332                                |       |
| 25 | [Java, ByteCode, Bound]                       | OF_RAW_TYPE              | [Type, File, Java, ByteCode, javaType]     | 3501                  | 7320                                |       |
| 26 | [Java, ByteCode, Bound, ParameterizedType]    | OF_RAW_TYPE              | [Type, File, Java, ByteCode, javaType]     | 3151                  | 7323                                |       |
| 27 | [Java, ByteCode, Bound, ParameterizedType]    | HAS_ACTUAL_TYPE_ARGUMENT | [Java, ByteCode, Bound, WildcardType]      | 2950                  | 7323                                |       |
| 28 | [Java, ByteCode, Parameter]                   | OF_GENERIC_TYPE          | [Java, ByteCode, Bound, ParameterizedType] | 2712                  | 13459                               |       |
| 29 | [Java, ByteCode, Member, Method, Constructor] | WRITES                   | [Java, ByteCode, Member, Field]            | 2585                  | 2177                                |       |

## Graph Density

```
total_number_of_nodes (vertices): 304000
total_number_of_relationships (edges): 968646
-> total directed graph density: 1.0481401341174772e-05
-> total directed graph density in percent: 0.0010481401341174772
```